



SGGS Institute of Engg. & Tech., Nanded
Dept. of Electronics & Telecom. Engg.
B. Tech. (EXTC) ISE-J Examination
Artificial Intelligence and Neural Networks (OEC-EC407)

Note: I. Solve ALL Questions
II. Assume suitable data if necessary

Date: 21 August 2025, Time : 11:00-11:45
Max marks: 20

Q.1 a) Discuss and derive the weight updating formula for **Delta learning rule**.

b) Find the weight vector of a neuron with $\tanh()$ activation function after ONE iterations of training using Learning Rule. The training data is:

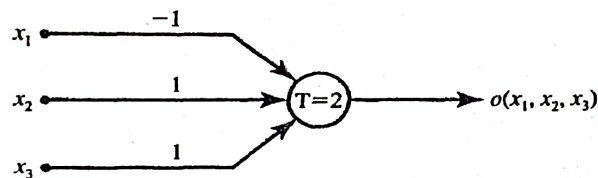
$X_1 = [-1, -3, -2, -1]^T$, $d_1 = '+1'$, $X_2 = [1, -3, 2, 1]^T$, $d_2 = '-1'$, $X_3 = [-1, -1, 2, 1]^T$, $d_3 = '-1'$

Assume learning constant is $c = 0.5$ and initial weight vector = $[-1, -2, 1, 1]^T$

(05)

Q.2 a) Explain McCulloch and Pitt's model of artificial Neuron (MPN). Find the logic implemented by following circuit of MPNs:

(05)



b) Perform Hebbian learning on a single-neuron. The task is to start with an initial weight vector

(05)

$$w^1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and use a learning constant $c = 1$ to update the weights using the following four input vectors:

$$x_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, x_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

You need to apply Hebbian learning steps sequentially to find the final weight vector after processing all four input vectors, using a bipolar binary activation function.

Course Outcomes (COs):

1. Understanding the basics of Artificial Intelligence, artificial neural network, their limitations, basic pattern analysis tasks such as classification and clustering, learning and adaptation using the learning rules, implementation of learning rule.
2. Describe the concepts of feed forward neural networks using single layer and multilayer networks to solve classification problem, and its implementation, single layer feedback networks to study the concept of memory using neural networks.
3. Analyze and implement the applications of neural networks in character recognition and control systems.
4. Understand Auto associative neural networks, Pattern storage and retrieval, Hopfield model, recurrent neural networks
5. Analyze Bayesian neural networks, Radial basis function networks
6. Understand self-organizing maps and recent trends in neural networks

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6- Create

Question No	CO No	BT No
1 (a,b)	1	1, 2
2 (a,b)	2	1, 6



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication
In-Semester Examination I
Academic Year: 2025-26 (Semester-I)

Class: Final Year B.Tech.	Course: Data Communication and Networking	Course Code: PEC-EC-403
Date: 19.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	All Questions are compulsory	Marks	CO	BT
Q.1	Explain Open Systems Interconnection (OSI) reference model?	05	CO1	BT2
Q.2	Write short notes on i) Routers ii) Switches.	05	CO2	BT1
Q.3	Explain Co-axial cable in detail?	05	CO2	BT1
Q.4	Compare circuit switching and packet switching?	05	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Describe the components and infrastructure that form the basis for most computer networks
CO 2	Describe the technical aspects of data communications on the Internet
CO 3	Achieve practical experience in network programming.
CO 4	Propose network designs based on case studies in colleges or other institutions
CO 5	Recognize security issues in a network, Design and Evaluate short path algorithm.